SG320HX System Solution

Technical Offer

SUNGROW

Version 1.0

EN

Technical Offer_SG320HX System Solution

Customer	Masdar & EDF-SA
Project Name	Amaala Solar PV Plant
Project Size	250MWac
Project Location	KSA
Version	V1.0

16 June 2022

SUNGROW

Contents

1. Introduction

This document describes the system design solution of SG320HX, including details of the design, selection of key equipment, etc.

1.1 Main application scenarios

SG320HX is mainly applicable to the power plant located at complex hill.

1.2 Design Principle

The design shall comply with existing national and industrial standards and pursue the lowest LCOE while meeting diversified demands of different customers.

Sungrow can provide matching 4.425MVA to 8.85MVA MV turnkey solution. The DC side can be flexibly configured based on factors such as solar site irradiance. This document does not cover specific projects which should be designed in accordance with customer requirements and plant types.

2. System Design

2.1 8.85MVA System SLD for SG320HX

The system mainly includes PV module arrays, string inverters, LV switch cabinet, transformer, switchgear and AC&DC cables. The power of the PV module array is converted by the inverter, then boosted by the transformer and finally fed into the grid. The following figure 1 shows the system topology.

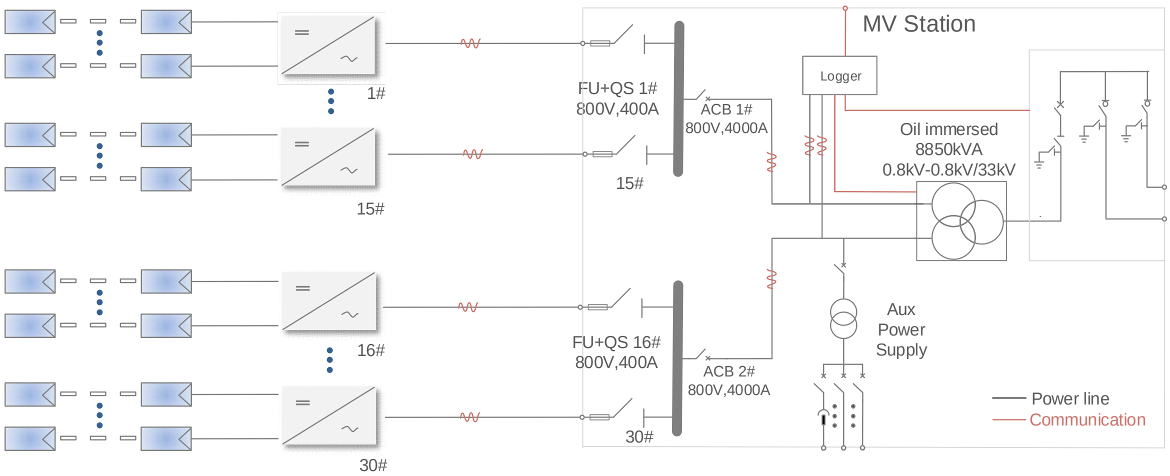


Fig.1. System Block Diagram

The system solution design includes PV module configuration, array layout, PV inverter, designs of LV switch cabinet, transformer, switchgear, cable and monitoring system, and other related system designs.

2.2 Sungrow Supply Scope

Sungrow scope of supply for Amaala Solar PV Plant in Saudi Arabia shown as table below(base on 50℃ design temperature, PF=0.974) and the nominal power output is 256.65MWac at 50℃ PF=1.

Table 2-1 Sungrow Scope of Supply for Amaala 250MW Project

No.	Items	Model	Unit	Qty.	Remark
1	**PV Inverter**	**SG320HX**	**unit**	**870**	**- Each PV Inverter consists of part 1.1.**
1.1	PV Inverter	SG320HX	unit	1	- IP66, 800Vac, 352kVA@30℃, 320kVA @40℃, 295kVA@50℃.
2	**MV Package**	**MVS8850-LV**	**set**	**29**	**- Each MV package consists of part 2.1~2.6.**
2.1	Enclosure	20 f.t. container	unit	1	- Standard 20HC f.t. container, RAL7035, C5
2.2	LV Cabinet	LV Cabinet	set	2	- 15 units of Fuse+Switch, 400A. - 1 unit of main ACB, 4000A. - Power meter/ MFM - IMD(Insulation monitoring)
2.3	MV Transformer	8850kVA	set	1	- Oil immersed, Dy11y11, 0.8kV-0.8kV /33kV,C5 - Rated 8850kVA @50C, 9600kVA @40C, 10560kVA @30C. - Efficiency 99% at full load pf=1. - Al/Al winding. - Gas relay & Winding temp indicator
2.4	MV Switchgear	RMU	set	1	- SF6 insulation, 630A, 36kV. - CCV, including 2 load breaker switch for ring connection, 1 vacuum circuit breaker for transformer protection. - Short circuit withstand: 20kA/3S. - 50/51, 50N/51N protection. - IAC AFL 20kA/1s - VCB Motor operation
2.5	Auxiliary	Aux. power	set	1	- Auxiliary transformer, Dry, 5kVA, Dyn11 - 3-phase, 800/400V. - 2kVA with 100W 2 hour VRLA battery for internal COM100 and RMU motor supply.
2.6	Monitoring Device	Block Communication	set	1	- Datalogger with PLC and PID module - 1 ethernet switch and fiber splice box
3	**PPC**	**Power Plant Controller**	**set**	**1**	**- Each PPC consists of part 3.1.**
3.1	PPC	PPC1000	set	1	- Server DELL R740 - Controller PPC100, PPC, Insight - Workstation Desktop computer - Firewall (Optional) - Panel 800 mm * 2260 mm * 600 mm

3. PV Inverter

3.1 SG320HX Inverter

SG320HX is a 1500VDC string inverter, and it's mainly applicable to PV power plant which need multi-MPPT feature or site location not easy to access.

Fig.2. SG320HX Inverter

Tab.1. SG320HX Characters

Values	Remark
High Yield	• 12 MPPTs with max. efficiency 99.01% • Compatible with 500Wp+ module • Data exchange with tracker system improving power generation
Lower BOS Cost	• Compatible with Al and Cu AC cables • Mach power line communication (MPLC) • Q at night function
Smart O&M	• Touch free commissioning and remote firmware upgrade • Smart IV curve diagnosis • Fuse free design with 2 strings per MPP
Proven Safety	• IP66 and C5 protection • Type II SPD for both DC and AC • Compliant with global safety and grid code

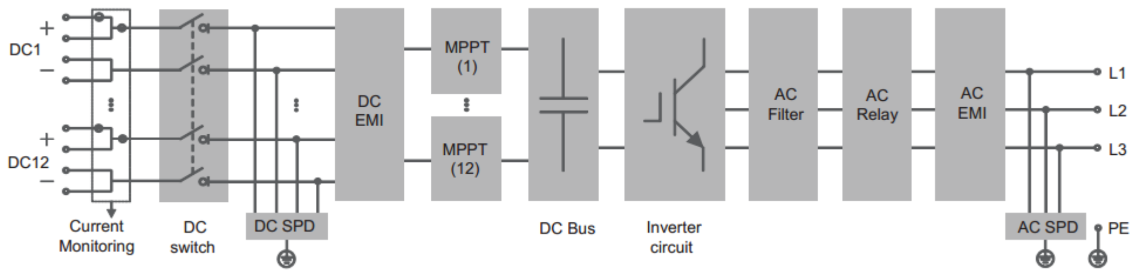


Fig.3. SG320HX Circuit Diagram

Tab.2.SG320HX Parameters

Specifications	Data
Input (DC)	
Max. PV input voltage	1500 V
Min. PV input voltage / Startup input voltage	500 V / 550 V
Nominal PV input voltage	1080 V
MPP voltage range	500 V – 1500 V
MPP voltage range for nominal power	860 V – 1300 V
No. of independent MPP inputs	12(optional: 14/16)
Max. number of input connectors per MPPT	2
Max. PV input current per MPPT	40 A (optional: 30 A for 14 or 16 MPP inputs)
Max. DC short-circuit current per MPPT	60 A
Output (AC)	
AC output power	352kVA@ 30℃ / 320kVA@ 40℃ / 295kVA@ 50℃
Max. AC output current	254 A
Nominal AC voltage	3 / PE, 800 V
AC voltage range	640 – 920V
Nominal grid frequency / Grid frequency range	50 Hz / 45 – 55 Hz, 60 Hz / 55 – 65 Hz
THD	< 3 % (at nominal power)
DC current injection	< 0.5 % In
Power factor at nominal power / Adjustable power factor	> 0.99 / 0.8 leading – 0.8 lagging
Feed-in phases / connection phases	3 / 3
Efficiency	
Max. efficiency	99.02 %
European efficiency	98.80 %
Protection	
DC reverse connection protection	Yes
AC short circuit protection	Yes
Leakage current protection	Yes
Grid monitoring	Yes
Ground fault monitoring	Yes
DC switch	Yes
AC switch	No
PV String current monitoring	Yes
Q at night function	Yes
Anti-PID and PID recovery function	Yes
Overvoltage protection	DC Type II / AC Type II
General Data	
Dimensions (W*H*D)	1136*870*361 mm
Weight	≤116 kg
Isolation method	Transformerless
Ingress protection rating	IP66
Night power consumption	< 6 W
Operating ambient temperature range	-30 to 60℃
Allowable relative humidity range (non-condensing)	0 – 100 %
Cooling method	Smart forced air cooling
Max. operating altitude	5000m (>4000m derating)
Display	LED, Bluetooth+APP
Communication	RS485 / PLC
DC connection type	MC4-Evo2 (Max. 6 mm² , optional 10 mm²)
AC connection type	OT/DT terminal (Max.400 mm²)
Compliance	IEC 62109, IEC 61727, IEC 62116, IEC 60068, IEC 61683, VDE-AR-N 4110:2018, VDE-AR-N 4120:2018, EN 50549-1/2, UNE 206007-1:2013, P.O.12.3, UTE C15-712-1:2013
Grid Support	Q at night function, LVRT, HVRT, active & reactive power control and power ramp rate control

3.2 Thermal Design

3.2.1 Temperature Characteristic Curves

Excessive increase in the internal temperature will shorten the life of electronic devices. Therefore, when the ambient temperature is high, the inverter will appropriately reduce the output power to ensure safe and long-term operation.

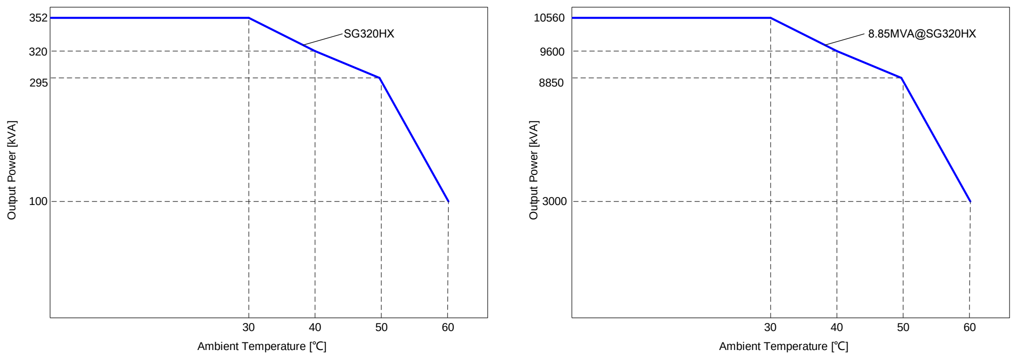


Fig.4. Inverter & Block Temperature Derating Curve

3.3 Grid Support

SG320HX provides sufficient reactive power compensation to meet the grid voltage support requirements. The power factor is 0.8 leading~0.8 lagging adjustable, and the power factor can be up to 0.9 at rated power. The P-Q curve is shown in figure 6. The ratio of reactive devices can be reduced on site.

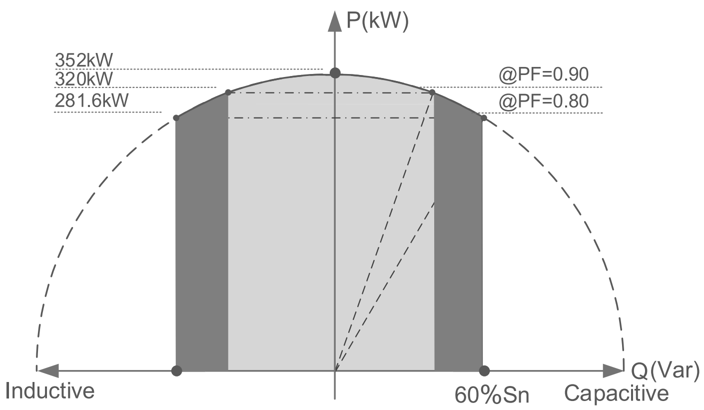


Fig.5. P-Q Curve

The maximum apparent power of SG320HX is 352kVA, and the maximum output reactive power is 211.2kVar.

3.4 PID Protection Design

3.4.1 Working Principle of PID Protection

The PID Module integrates with anti-PID function and PID recovery function.

Anti-PID function is to raise the PV- to ground voltage of P-type solar panel to above 0V, and the PV+ to ground voltage of N-type solar panel to below 0V during the daytime, so as to prevent PID effect. PID recovery function is to raise the PV- to ground voltage of P-type solar panel to the target voltage, and the PV+ to ground voltage of N-type solar panel to the negative value of target voltage at night, so as to suppress the PID effect. The default target voltage is 500V (which can be set through App).

3.4.2 PID Work Mode

Anti-PID function and PID recovery function both need to be enabled through the App.

① Anti-PID mode:

- Turn on Anti-PID;
- Communication works normally;
- Active power greater than 5% of rated power;
- Bus voltage is less than 1350V;
- P-type solar panel: 100V(±20V) < PV+ to ground voltage < 1350V(±50V); N-type solar panel: -1350V(±50V) < PV- to ground voltage < -100V(±20V);
- The delay time is reached, and the delay time can be set from 1 ~ 3600s (default 3600s).

When the above conditions are met at the same time, Anti-PID is enabled.

② PID recovery mode:

- Turn on PID recovery;
- Communication interruption;
- Perform within the allowable time period, the allowable time period is between 22:00-05:00, the allowable time period can be set;
- The voltage difference between PV+ and PV- is less than 150V;
- P-type solar panel: PV- to ground voltage < 1350V; N-type solar panel: PV+ to ground voltage > -1350V.

When the above conditions are met at the same time, PID recovery is enabled.

3.4.3 Requirements of Anti-PID Mode for Transformer & AC cable

①If the LV-side winding is in Y type, grounding of neutral point is prohibited.

②The AC SPD of the LV-side winding to ground should be more than 906V.

③The LV-side winding of the transformer, AC cables, and secondary devices (including protective relay, detection and measurement instruments, and related auxiliary devices) must withstand at least 906V voltage to ground.**AC cable insulation voltage must choose 1.8/3KV**

4. MV Station

The MV station integrates LV cabinet, MV transformer, MV switchgear, communication box and aux. power supply.

Fig.6. 8.85MVA MV Station

4.1 MV Station System Single Line Diagram

The LV-side voltage of the transformer must match with the AC output of the inverter, and the HV-side voltage should be designed according to the voltage at grid-connected point. The transformer capacity depends on the capacity of the block while the circuit breaker depends on the output of the inverter. The rated output voltage of SG320HX is 800V, and the frequency is 50Hz/60Hz. Figure 8 shows the system single line diagram by using 8.85MVA block as an example in which the grid-connection voltage is 33kV.

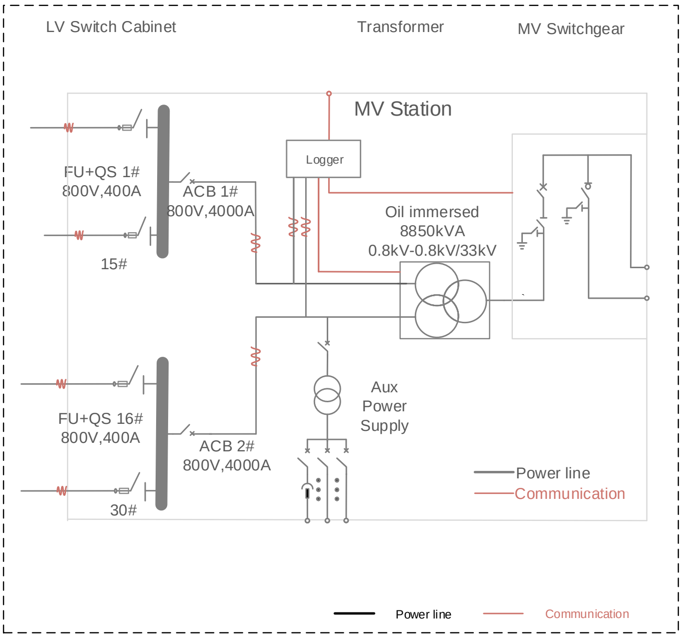
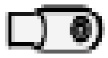


Fig.7. MV Station System SLD

4.2 LV Switch Cabinet

The LV switch cabinet must be used downstream of the inverters for protection and maintenance purposes. The LV switch cabinet is integrated into the MV station as its LV compartment.

Item	Model	Descriptions	Remarks
Main Switch	Load switch	Ue=800V,In=4000A	Qty: 2
Branch Switch	FU+QS	Ue=800V, In=400A	Qty:30
Input Cable Terminal	OT/DT Terminal	Cable selection (per polarity): Al or Cu: Max. 300mm^2(591.7kcmil)	
Cable entry	Bottom entry	Bottom entry	
Enclosure	Non-walk-in, outdoor	IP20	

4.3 MV Transformer

It is recommended to use 8.85MVA transformer. Typical design and design considerations are described below. The transformer type is an oil-immersed and suitable for outdoor installation.

The LV bushings are connected to the AC side of the inverter via cables, and the MV bushings are connected to the MV switchgear via MV screened separable cable connectors, all for outdoor installation.

Safety barrier with warning sign is placed at the entrance of transformer compartment for preventing from entering hazardous area unintentionally.

Oil immersed distribution transformer, suitable for outdoor installation, insulation class A

Items	Specifications
Type	Oil immersed
Rated power	8850kVA@50℃
Overload	10560@30℃
Insulation class	A
Frequency	60Hz
Number of phase	3
Vector group	Dy11y11
LV / MV voltage	0.8kV-0.8kV / 33kV
Tapping on HV	0, ±2x2.5%
Impedance	9.5% (±10%)
Material of winding(HV/LV)	Al/Al (Cu/Cu optional)
Cooling method	ONAN
IP protection	Transformer body:IP68, Other parts: IP54
Highest system voltage	36kV
Power frequency withstand voltage	70kV
Rated lightning impulse withstand voltage	170kV
Secondary insulation level	LI-/AC3kV
Transformer oil type	Mineral oil(PCB free)
Temp rise (winding/top oil)	55/50 K@50℃

Painting color	RAL7035 (other color optional)
Site Conditions	
Ambient temperature	-20~50℃（No derating）
Altitude	1000m (>1000m optional)
Accessories	
LV bushings	6 piece
HV bushings	3 piece
HV connector	3 piece
Cable termination	1 piece
Off-load tap changer	1 piece
Oil filling device	1 piece
Pressure relief valve	1 piece
Oil drain valve	1 piece
Thermometer pocket	1 piece
Oil-temperature indicator	1 piece
Oil-level indicator	1 piece
Earthing terminal	1 piece
Signals	
Oil-temperature signal	1 piece
Oil-temperature alarm	1 piece
Oil-temperature trip	1 piece
Pressure relief trip	1 piece
Oil level trip	1 piece
Light gas alarm	1 piece
Heavy gas trip	1 piece
Winding temperature alarm	1 piece
Winding temperature trip	1 piece
Tests	
Routine test as per IEC60076	Included for all unit
Type test as per IEC60076	Type test report or only one unit applied
Special test as per IEC60076	Optional

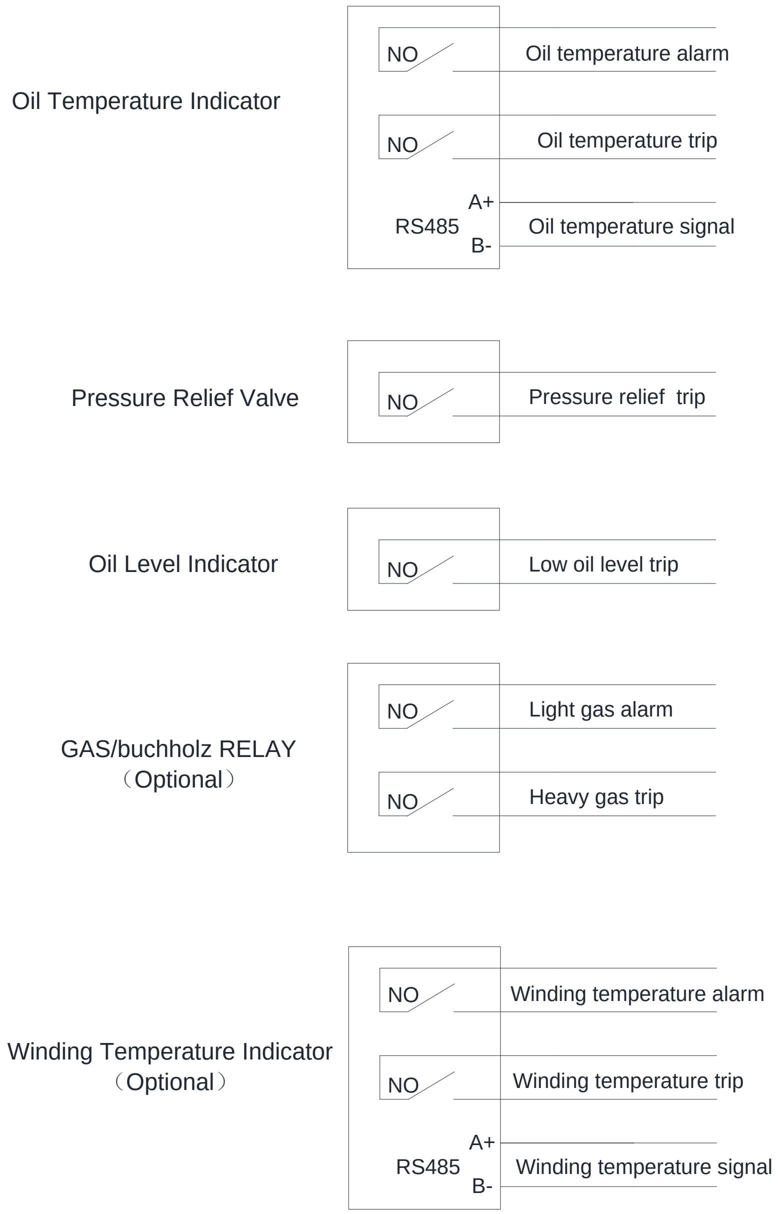


Fig.8. Transformer Signal Terminals

Transformer requirement

1) The transformer must be suitable for operating with a pulsed inverter.

2) In case anti-PID solution (Potential Induced Degradation) is used, transformer secondary windings shall be able to handle the AC phase voltage to ground voltage Vrms=906V ($\sqrt{V_{AN}^2 + V_{Ng}^2}$) for the 1500V system. The LV side of the transformer and the maximum continuous voltage of lightning protection devices for the LV side of the transformer should comply with this requirement.

3) The output voltage of SG320HX is 800Vac L-L (640~920Vac), 50Hz or 60Hz. The voltage at the MV side shall be in line with the MV grid voltage at the connection point, transformer with tap changer is preferred.

4) Transformer capacity shall be in line with the inverters max. output power.

5) Double split (two for LV, one for MV side) transformer is to be used, Dy11y11 connections can be used. Double winding (one for LV, one for MV side) transformer is to be used, Dy11 connections can be used.

6) For double split transformer, the LV1-LV2 impedance is recommended to be no less than 8% (based on the half of the nominal transformer kVA rating).

7) The transformer shall be capable of withstanding a certain level of harmonic current, the max. total harmonic current is 5% of fundamental current at nominal power.

8) The transformer shall be able to withstand a certain level of DC component, 0.5% of fundamental current at nominal power.

9) The transformer shall be able to withstand a certain phase imbalance, 5% of the current at nominal power.

10) The LV winding of the MV transformer shall withstand a voltage gradient du/dt of up to 500V/μs to ground.

11) The LV winding of the MV transformer shall withstand a current gradient di/dt of up to 150A/ms.

12) Electrostatic shield and isolation between LV and MV windings recommended.

13) The transformer shall be grounded at only one point, and the grounding of the transformer must be secure.

14) The insulation class shall be at least F for dry type transformer, A for oil type.

Insulation class	A	E	B	F	H	N	C
Max. temp. (℃)	105	120	130	155	180	200	220
Temp. raise of winding (K)	60	75	80	100	125	135	150

15) The load weighed efficiency shall be compliant with the latest local requirements.

16) The local environment (temperature, humidity, altitude, air etc.) shall be taken into consideration.

17) The temperature raise and min. electric clearance shall be taken into consideration especially when the altitude is over 1000m.

18) The country specific grid frequency & standards shall be respected.

4.4 MV Switchgear

MV switchgear is the combination of electrical disconnect switches, fuses or circuit breakers used to control, protect and isolate electrical equipment. Switchgears are used both to de-energize equipment to allow work to be done and to clear faults downstream. This type of equipment is directly linked to the reliability of the electricity supply. MV switchgear includes two load breaker switch, one vacuum circuit-breaker for transformer protection.

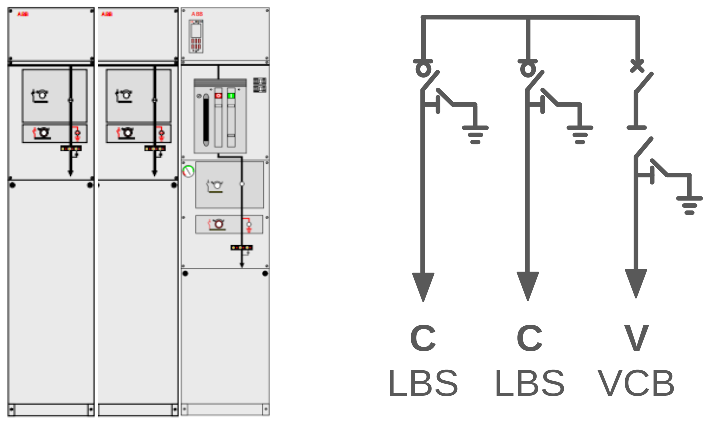


Fig.9. Single Line Diagram

SafeRing 36		C-module		F-module		V-module	
		Switch disconnector	Earthing switch	Switch-fuse disconnector	Downstream earthing switch	Vacuum circuit-breaker	Earthing switch/ disconnector
Rated voltage	kV	36/38,5/40,5	36/38,5/40,5	36/38,5/40,5	36/38,5/40,5	36/38,5/40,5	36/38,5/40,5
Power frequency withstand voltage	kV	70/80/95	70/80/95	70/80/95	70/80/95	70/80/95	70/80/95
- across disconnector	kV	80/95/118		80/95/118			80/95/118
Lightning impulse withstand voltage	kV	170/180/185	170/180/185	170/180/185	170/180/185	170/180/185	170/180/185
- across disconnector	kV	195/210/215		195/210/215			195/210/215
Rated normal current	A	630/630/630[1)]		200/200/200[2)]		630/630/630[1)]	
Breaking capacities:							
- active load	A	630/630/630		200/200/200			
- closed loop	A	630/630/630		200/200/200			
- off load cable charging	A	20/21/21		20/21/21		50/50/50 (Class C1)	
- earth fault	A	60/63/63		60/63/63			
- earth fault cable charging	A	35/36/36		35/36/36			
- transfer current	A			840/750/750			
- short-circuit breaking current	kA			see[3)]		20/20/20	
						(Class E1,S1)	
Making capacity	kA	50/50/50 (5 times)	50/50/50 (5 times)	see[3)]	2,5/2,5/2,5 (5 times)	50/50/50	50/50/50
Class (Electrical endurance)		E3/E2/E2	E2/E2/E2	- / -	E2/E2/E2	E1/E1/E1	E2/E2/E2
Short time current 1 sec. [4)]	kA	25/25/25	25/25/25		1/1/1	25/25/25	25/25/25
Short time current 3 sec.	kA	20/20/20	20/20/20	see[3)]		20/20/20	20/20/20
Internal arc classification IAC AFL, 1s	kA	20/20/20		20/20/20		20/20/20	

No.	Items	Option
1	Motor operation for vacuum circuit-breaker	Optional, included in the offer
2	MV cable connector for Load break switch panel	Optional, excluded If customer want Sungrow supply , customer should provide cable specs before signing PO with extra cost
3	MV Surge Arrester for Load break switch panel	Optional, excluded If customer want Sungrow supply , customer should provide cable specs before signing PO with extra cost and cable connector for load break switch panel shall be together for SUNGROW supply to ensure they are compatible with each other.
4	MV Surge Arrester for VCB panel	Optional, included

5	Internal arc classification(IAC)	AFL 20kA/1S
6	VCB Relay Protection	50/51, 50N/51N
7	Altitude	Standard offer is 1000m, (>1000m optional)

4.5 Interlock

RMU internal interlock between circuit breaker, switch disconnector, earthing switch and cable compartment for each module as per IEC62271-200 required

5. Monitoring System Design

5.1 Block Communication Design

Block Communication is used in the 8.85MVA block to achieve data collection and transmission. The Block Communication mainly includes a Logger, two master PLC communication modules and an optical fiber switch. The inverter communicates with the Logger via RS485 cables or the PLC communication module. Devices such as the transformer, meteo station, and energy meter communicate with the Logger through RS485, DI/DO signal or analog signal. The Block Communication will transfer the information to the power plant monitoring system via the optical fiber ring network by IEC104 or Modbus TCP protocol. The communication solution is shown in the figure 12.

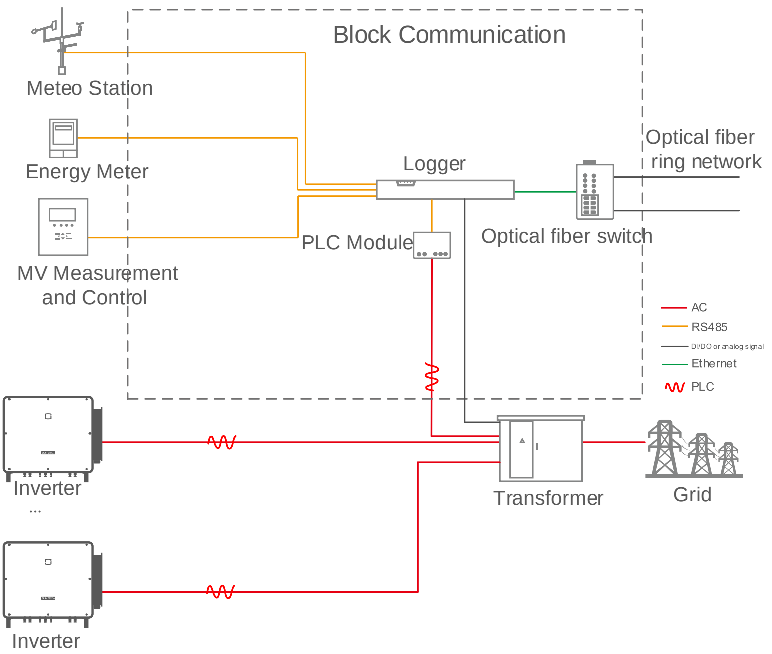


Fig.10. SG320HX Block Communication (PLC communication solution)

5.2 Communication Networking

Figure 13 shows the plant networking and monitoring topology, mainly including block communication, local monitoring, remote monitoring iSolarCloud, etc.

The Block Communication (integrating with Logger3000) achieves communication and monitoring of a block, each block forms a ring network through optical fiber.

Each block information is uploaded to the local monitoring system via optical fiber, and then the local monitoring system uploads the information to cloud computing platform via isolating device. In this way, iSolarCloud can centrally monitor and manage the plant.

iSolarCloud integrates operation management platform and maintenance management platform. The operation management platform can monitor and analyse data of all plants monritored by it and provides guidance on plant operation based on multi-dimensioned operation indexes. The maintenance management platform can remotely monitor core plant devices, quickly locate the faulty device, improve O&M efficiency, and reduce O&M costs. In addition, iSolarCloud supports access of third-party devices and system platforms.

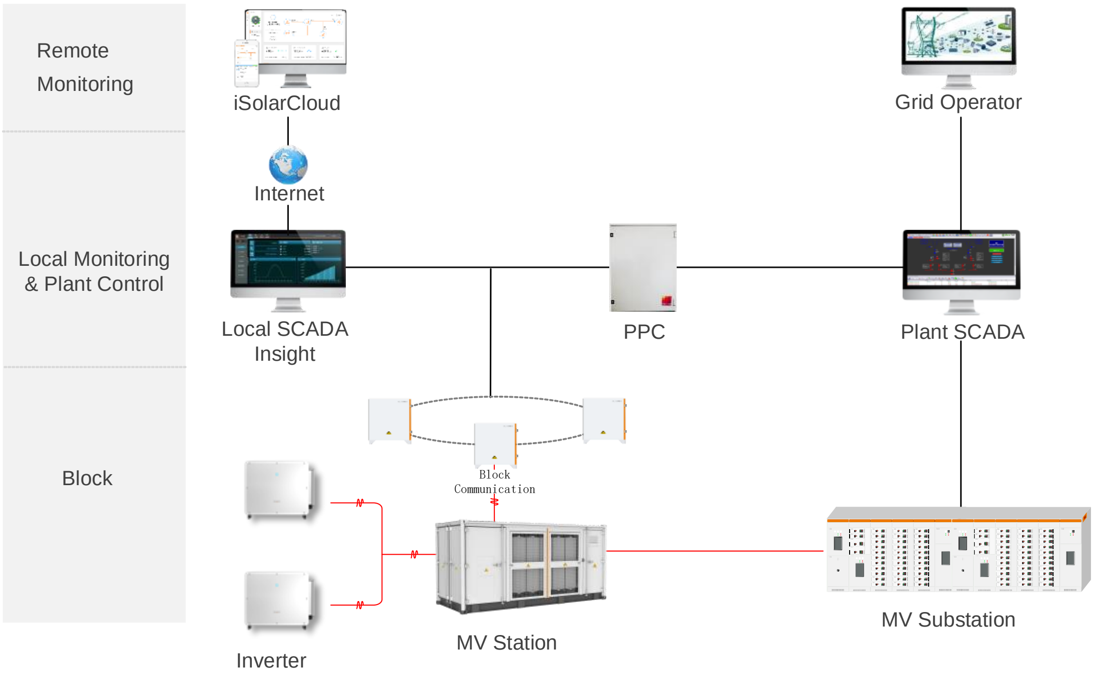


Fig.11. Plant Networking and Monitoring Typology

Sungrow provides Block Communication，Insight，iSolarCloud and PPC. For details, refer to related documents.

6. Annexure

6.1 Standards and Codes

Item	Name
Safety	IEC 62109-1: Safety of power converters for use in photovoltaic power systems,part1: General requirements
	IEC 62109-2: Safety of power converters for use in photovoltaic power systems,part1: General requirements, part 2: Particular requirements for inverters
Grid Connection	IEC 61727: Photovoltaic (PV) systems - Characteristics of the utility interface
Design Requirements /References	IEC62548: Photovoltaic (PV) arrays – Design requirements
	IEC60364-1: Low voltage electrical installations-Part 1: Fundamental principles, assessment of general characteristics, Definitions IEC 60364-4-43_2008 Protection for safety – Protection against overcurrent IEC 60364-7-712, Electrical installations of buildings - Part 7-712: Requirements for special installations or locations - Solar photovoltaic (PV) power supply systems
	IEC62271-202High-voltage/low-voltage prefabricated substation
	IEC 60947-2: Low-voltage switchgear and control gear, Part 2: Circuit breakers
	IEC 62548: Photovoltaic (PV) arrays – Design requirements
	IEC60287-3-1: Electric cables –Calculation of the current rating – Part 3-1:Sections on operating conditions –Reference operating conditions and selection of cable type
	IEC 60269-1 – Low-voltage fuses – Part 1: General requirements IEC 60269-2 – Low-voltage fuses – Part 2: Supplementary requirements for fuses for use by authorized persons (fuses mainly for industrial application) IEC60269-6: Low-voltage fuses – Part 6: Supplementary requirements for fuse-links for the protection of solar photovoltaic energy systems;
	IEC 61643-11: Low-voltage surge protective devices - Part 11: Surge protective devices connected to low-voltage power systems

6.2 Contact Information

For any queries about this product, please contact us on below.

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